

B.Sc. (Entire Computer Science)-I, Level - 4.5 UG Certificate Level							
Sem:		I					
Paper Category:		DSC2-1 (Major)					
Paper Name:		Python Programming-I			Paper Code : G06-0102		
Credit:		02		Theory:		2 Hrs./Week	
Marks:	UA:	30	CA:	20	Total:	50	

Course Objective:

1. To learn the fundamentals of Python programming
2. To learn different data structures used in Python
3. To learn different control statements used in logic development.
4. To learn the various operations on the array, list, tuple, string, set, and dictionary.

Course Outcomes:

Upon successful completion of this course, students will be able to

1. Understand the basic concepts and applications of Python.
2. Design, create, build, and debug Python applications.
3. Explore the Integrated Development Environment (IDE).
4. Write and apply decision structures for different operations.
5. Write loop structures to perform iterative tasks.

Unit-I: Basics of Python:	No. of Lectures:12	Weightage: 9-12 Marks
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Introduction: Features of Python, Python Virtual Machine, Memory management, Garbage Collection, Installation of Python, setting the path to operating system environment, writing the first Python program, executing a Python program.

Datatypes in Python: Datatypes-Numeric, Sequence Type-String, List, Tuple, Boolean, Set, Dictionary, Binary Types, Type conversion- implicit and explicit, Python comments, literals, constants, Identifiers, naming conventions, operators, operator precedence and associativity, input and output statements, command-line arguments.

Control Statements: if statement, if..else statement, if..elif..else statement, while loop, for loop, else suite, infinite loop, nested loops, word indentation, break statement, continue statement, pass statement, assert statement, return statement.

Unit-II: Data Structure and Function:	No. of Lectures:18	Weightage: 18-21 Marks
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String, List, Tuple, Set and Dictionary: Creating string, manipulating different operations on string, creating list, manipulating different operations on list, list comprehensions, creating tuple, manipulating different operations on tuple, creating set, manipulating different operations on set, creating dictionary, manipulating different operations on dictionary.

Functions: Difference between function and method, defining a function, calling function, returning result from a function, returning multiple values from a function, functions are objects, formal and actual arguments, types of arguments, local, nonlocal and global variables, global keyword, recursive functions, anonymous functions or lambdas, using lambdas with filter(), map() and reduce() functions

Arrays in Python: Introduction, advantages of array, creating an array, types of arrays, importing array module, indexing and slicing on arrays, methods of array module.

Reference Books:

1. Python: The Complete Reference by Martin C. Brown.
2. Core Python Programming, Dreamtech publications, by R. Nageswara Rao.
3. Python Programming, A modular approach, First Edition, Pearson, by Taneja Sheetal
4. Learning with Python, Dreamtech publications, by Allen Downey
5. Python Programming for the Absolute Beginner by Michael Dawson-Cengage Learning.

B.Sc. (Entire Computer Science)-I, Level - 4.5 UG Certificate Level							
Sem:		I					
Paper Category:		Practical (Major)					
Paper Name:		Practical based on DSC2-1			Paper Code : G06-0102-P		
Credit:		02		Practical:		4 Hrs./Week	
Marks:	UA:	30	CA:	20	Total:	50	

1. Write a simple Python script that prints "Hello, World!" to the console.
2. Perform basic arithmetic operations (addition, subtraction, multiplication, division) on numeric variables.
3. Concatenate strings and convert between data types as needed.
4. Implement conditional statements (if-else, elif) to control program flow based on different conditions.
5. Use loops (for, while) to iterate over lists, tuples, dictionaries, and ranges.
6. Create nested loops and conditional statements for more complex control flow logic.
7. Create and manipulate lists, tuples, dictionaries, and sets.
8. Access elements in data structures using indexing and slicing operations.
9. Use list comprehensions and generator expressions for concise data manipulation.
10. Write a Python program to find the sum of a list of numbers using a for loop.
11. Write a Python program to display stars in right angled triangular form using nestedfor loops.
12. Write a Python program to display a multiplication table from 1 to 10 using nested for loops.
13. Write a Python program to display elements in a list in reverse order.
14. Write a Python program to accept elements in the form of a tuple and display theirsum and average.
15. Write a Python program to create a dictionary with employee details and retrieve the values upon giving keys.